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Oefeningenreeks 5G nr. 6.2

$$f(x) = \sqrt{x}$$

$$g(x) = 2\sqrt[4]{x}$$

$$f(x) = g(x)$$

$$\sqrt{x} = 2\sqrt[4]{x}$$

$$\Leftrightarrow x^{\frac{1}{2}} = 2x^{\frac{1}{4}}$$

$$\Leftrightarrow x^{\frac{1}{2}} - 2x^{\frac{1}{4}} = 0$$

$$\Leftrightarrow x^{\frac{1}{4}} \left(x^{\frac{1}{4}} - 2 \right) = 0$$

$$\Leftrightarrow x^{\frac{1}{4}} = 0 \text{ of } x^{\frac{1}{4}} = 2$$

$$\Leftrightarrow x = 0 \text{ of } x = 16$$

$$\Rightarrow x \in [0, 16]$$

We testen een punt tussen 0 en 16: $x = 1$

$$f(1) = \sqrt{1} = 1$$

$$g(1) = 2\sqrt[4]{1} = 2$$

$\Rightarrow g(x)$ is de bovenste functie

$$\Rightarrow V = \pi \int_0^{16} (g(x)^2 - f(x)^2) dx$$

$$= \pi \int_0^{16} ((2\sqrt[4]{x})^2 - (\sqrt{x})^2) dx$$

$$(2\sqrt[4]{x})^2 = 4x^{\frac{1}{2}}$$

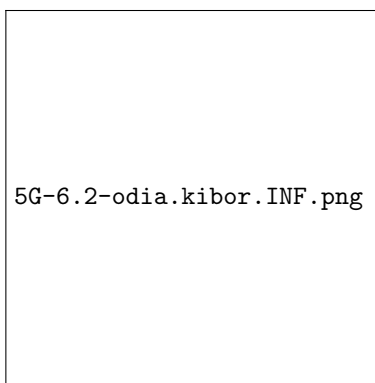
$$(\sqrt{x})^2 = x$$

$$\Rightarrow (2\sqrt[4]{x})^2 - (\sqrt{x})^2 = 4x^{\frac{1}{2}} - x$$

$$\Rightarrow V = \pi \int_0^{16} (4x^{\frac{1}{2}} - x) dx$$

$$= \pi \left[\frac{4x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^2}{2} \right]_0^{16}$$

$$\begin{aligned}
&= \pi \left[\frac{8}{3} x^{\frac{3}{2}} - \frac{x^2}{2} \right]_0^{16} \\
&= \pi \left(\frac{8}{3} (16)^{\frac{3}{2}} - \frac{16^2}{2} \right) - \pi \left(\frac{8}{3} (0)^{\frac{3}{2}} - \frac{0^2}{2} \right) \\
16^{\frac{3}{2}} &= (\sqrt{16})^3 = 4^3 = 64 \\
\Rightarrow V &= \pi \left(\frac{8}{3} \cdot 64 - \frac{256}{2} \right) - \pi(0) \\
&= \pi \left(\frac{512}{3} - 128 \right) \\
&= \pi \left(\frac{512}{3} - \frac{384}{3} \right) \\
&= \pi \cdot \frac{128}{3} \\
\Rightarrow V &= \frac{128\pi}{3}
\end{aligned}$$



References